

Lab Report 1: Introduction to Python, NumPy, OpenCV, and Matplotlib

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Objective

The goal of this lab is to introduce students to fundamental programming concepts in Python, array manipulation using NumPy, and image processing using OpenCV and Matplotlib. This also includes understanding color spaces like BGR, RGB, and HSV.

Tools and Libraries Used

- Python 3.x
- NumPy
- OpenCV
- Matplotlib

Code Explanation

1. Python Basics: Loops and Arrays

```
1 for i in range(5):
2     print(f"Loop iteration: {i}")
3
4 array = np.array([1, 2, 3, 4, 5])
5 print("NumPy Array:", array)
6
7 def square_elements(arr):
8     return np.square(arr)
9
```

```

10 squared_array = square_elements(array)
11 print("Squared Array:", squared_array)

```

Listing 1: Python Loop and NumPy Array

2. Reading and Displaying Image (BGR vs RGB)

```

1 image_bgr = cv2.imread('images/image1.jpg')
2 plt.subplot(1, 2, 1)
3 plt.imshow(image_bgr)

```

Listing 2: Reading and Displaying Image

Note: OpenCV reads images in BGR color space, which causes colors to appear swapped when displayed using Matplotlib (which uses RGB).

3. Color Space Conversion and HSV Display

```

1 image_hsv = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2HSV)
2
3 cv2.imshow('HSV Image', image_hsv)
4 cv2.imwrite('output_image_hsv.jpg', image_hsv)
5 cv2.waitKey(0)
6 cv2.destroyAllWindows()
7
8 plt.subplot(1, 2, 2)
9 plt.title('HSV Image in Matplotlib')
10 plt.imshow(image_hsv)
11 plt.show()

```

Listing 3: HSV Conversion and Display

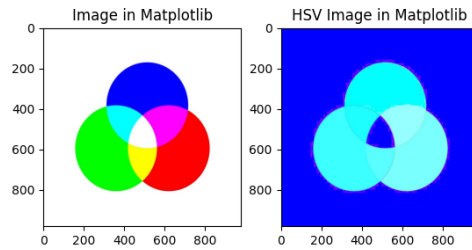
Explanation: The image is converted from BGR to HSV. OpenCV is used to display and save the image in HSV format. The HSV image is also plotted using Matplotlib for comparison.

Results and Observations

- Images displayed using OpenCV appear different from those shown using Matplotlib due to BGR vs RGB format mismatch.
- Converting to HSV provides better segmentation and color filtering options.
- NumPy allows for fast array operations and transformations.

Discussion: BGR vs RGB

OpenCV uses BGR by default because early standards and camera drivers supported this format more efficiently. Matplotlib, being visualization-focused, follows the RGB convention which aligns with most image viewers and standards.



(a) Image displayed using OpenCV (BGR)

This leads to color distortion when an OpenCV BGR image is directly shown in Matplotlib without conversion.

Conclusion

This lab provided an introduction to Python programming and image processing with OpenCV and Matplotlib. Students learned how to handle loops, arrays, functions, and color space conversions. Understanding the difference between BGR and RGB is crucial when working across different image processing libraries.

References

- OpenCV Documentation: <https://docs.opencv.org/>
- Matplotlib Documentation: <https://matplotlib.org/>
- NumPy Documentation: <https://numpy.org/>